Advanced Quantum Mechanics - Comprehensive Quiz

Based on TU Darmstadt Lecture Material 2025/2026

Instructions

This quiz covers: Fundamentals (F), Scattering Theory (S), Many Body Physics (M), and Relativistic Quantum Mechanics (R). Each section contains 5 blocks with 5 statements each. Statements are either **True** or **False**. The number of correct statements is indicated for each block.

1 Fundamentals of Quantum Mechanics (F)

Block F1: Schrödinger Equation and Dirac Notation

Number of correct statements: 4

i. The state vector $|\psi(t)\rangle$ lives in a complex Hilbert space and contains all physical information.

ii. The inner product $|\langle\phi|\psi\rangle|^2$ represents the probability of finding a system in state $|\phi\rangle$.

iii. For an operator to represent a measurable observable, it must be Unitary.

iv. The TDSE ensures that the norm of the state is preserved if the Hamiltonian is Hermitian.

v. In Dirac notation, a bra vector $\langle\psi|$ corresponds to the Hermitian conjugate of the ket $|\psi\rangle$.

Block F2: Time Evolution and Dyson Series

Number of correct statements: 3

i. For a time-independent Hamiltonian, the evolution operator is $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$.

ii. If $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$, the solution is a simple exponential of the time-integrated Hamiltonian.

iii. The time-ordering operator $\mathcal{T}$ ensures earlier times act to the left of later times.

iv. The Dyson expansion is a power series representation of the time-ordered exponential.

v. In the energy eigenbasis, time evolution attaches a phase factor $e^{-iE_n t/\hbar}$ to each component.

Block F3: Two-Level Systems (Qubits)

Number of correct statements: 4

i. A Hamiltonian $H = \hbar\Omega\sigma_x$ can cause transitions between $|0\rangle$ and $|1\rangle$.

ii. The eigenvalues of σ_x are ± 1, leading to energy levels $\pm\hbar\Omega$.

iii. The state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is an eigenvector of σ_z.

iv. Time evolution of $|0\rangle$ under $H = \hbar\Omega\sigma_x$ gives $|\psi(t)\rangle = \cos(\Omega t)|0\rangle - i\sin(\Omega t)|1\rangle$.

v. σ_x is its own inverse and is also Hermitian.

Block F4: Position and Momentum Representations

Number of correct statements: 2

i. The momentum operator in position representation is $\hat{p} \to -i\hbar\nabla$.

ii. The completeness relation $\int dx|x\rangle\langle x| = 1$ allows defining $\psi(x) = \langle x|\psi\rangle$.

iii. The position operator $\hat{x}$ acts as a derivative in the position representation.

iv. The commutator $[\hat{x}, \hat{p}]$ is equal to zero in the position representation.

v. Momentum eigenstates are localized Delta functions in position space.

Block F5: The Free Particle

Number of correct statements: 3

i. For a free particle ($V(x) = 0$), the energy spectrum is continuous and positive ($E > 0$).

ii. Plane wave solutions $\phi_k(x) = \frac{1}{\sqrt{2\pi}}e^{ikx}$ are square-integrable (L^2).

iii. The dispersion relation for a non-relativistic free particle is $E_k = \frac{\hbar^2 k^2}{2m}$.

iv. Any initial state $\psi(x, 0)$ can be evolved by applying time-dependent phases to its Fourier components.

v. Negative energy solutions for a free particle are standard bound states in non-relativistic QM.

2 Scattering Theory (S)

Block S1: Basics and Cross-Sections

Number of correct statements: 4

i. The scattering amplitude $f(\theta, \phi)$ relates theory to experimental cross-sections.

ii. The differential cross-section is defined as $d\sigma/d\Omega = |f(\theta, \phi)|^2$.

iii. Scattering is used to probe the shape and strength of interaction potentials.

iv. Integrating the differential cross-section $|f|^2$ over the full solid angle yields the total cross-section.

v. The scattering amplitude f has the physical dimensions of energy.

Block S2: Lippmann-Schwinger Equation

Number of correct statements: 3

i. The Lippmann-Schwinger equation is an integral form of the Schrödinger equation.

ii. It describes $|\psi^{(+)}\rangle$ as a sum of the incident plane wave and a scattered term.

iii. The Green's function $\hat{G}(E_i)$ uses an $i\epsilon$ term to ensure outgoing wave boundary conditions.

iv. The Lippmann-Schwinger equation is only valid for potentials stronger than the kinetic energy.

v. In the momentum basis, the free Green's function operator is non-diagonal.

Block S3: Born Approximation

Number of correct statements: 3

i. The first Born approximation replaces the full state with the incident plane wave in the integral.

ii. The Born approximation is most accurate for very strong potentials at low energies.

iii. For a spherically symmetric potential, the Born amplitude depends only on the momentum transfer q.

iv. The Born amplitude is proportional to the Fourier transform of the potential $V(r)$.

v. The Born approximation always strictly preserves the unitarity of the S-matrix.

Block S4: Partial Wave Expansion

Number of correct statements: 4

i. Partial wave analysis decomposes scattering into components of definite angular momentum l.

ii. At low energies ($ka \ll 1$), the s-wave ($l = 0$) usually dominates the scattering.

iii. The phase shift δ_l characterizes the shift of the radial wavefunction's nodes due to $V(r)$.

iv. If a phase shift δ_l is zero, there is no scattering in that specific l-channel.

v. For a hard sphere, all phase shifts δ_l are always positive.

Block S5: Low Energy Limits (Hard Sphere)

Number of correct statements: 2

i. In the low-energy limit, scattering from short-range potentials becomes isotropic.

ii. The scattering length a is defined by the $k \to 0$ limit of the s-wave amplitude.

iii. The Born approximation is always exact for a hard sphere potential at any energy.

iv. For a square-well potential, scattering is always anisotropic as $k \to 0$.

v. The total cross-section σ at $k \to 0$ for a hard sphere of radius R is exactly πR^2.

3 Many Body Physics (M)

Block M1: Indistinguishable Particles

Number of correct statements: 3

i. Distinguishable particles require wavefunctions to be totally symmetric.

ii. Bosons have symmetric wavefunctions and follow Bose-Einstein statistics.

iii. Fermions follow the Pauli Exclusion Principle (antisymmetric wavefunctions).

iv. Exchange symmetry of a many-body wavefunction has no effect on physical observables.

v. Indistinguishability is a fundamental property of identical particles like electrons.

Block M2: Second Quantization - Operators

Number of correct statements: 4

i. Bosonic creation/annihilation operators satisfy $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$.

ii. Fermionic operators satisfy anti-commutation relations $\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij}$.

iii. For fermions, $(\hat{c}_j^\dagger)^2 = 0$, representing the Pauli principle.

iv. The number operator $\hat{n}_j = \hat{c}_j^\dagger \hat{c}_j$ counts particles in state j.

v. Applying an annihilation operator to the vacuum state $|0\rangle$ yields $|0\rangle$.

Block M3: Fock States and Distributions

Number of correct statements: 3

i. A Fock state $|n_0, n_1, ...\rangle$ is an eigenstate of the mode number operators.

ii. In second quantization, the vacuum state $|0\rangle$ is the state with zero particles.

iii. A thermal state is a coherent superposition with a single fixed phase.

iv. For bosons, any number of particles can occupy the same mode j.

v. The mean occupation number in a thermal state follows the Bose-Einstein distribution for bosons.

Block M4: Gross-Pitaevskii and Hubbard Models

Number of correct statements: 4

i. The Gross-Pitaevskii equation (GPE) is a non-linear Schrödinger equation for BECs.

ii. The GPE is derived by taking the expectation value of the Heisenberg field equations.

iii. The Hubbard model describes hopping (t) and on-site interactions (U) on a lattice.

iv. In the limit $U \gg t$ at half-filling, the Hubbard model maps to the Heisenberg spin model.

v. The Hubbard model is only applicable to non-interacting bosonic systems.

Block M5: Magnetism and Spin Models

Number of correct statements: 2

i. The Heisenberg model describes interactions between neighboring spins $\mathbf{S}_i \cdot \mathbf{S}_j$.

ii. The Ising model is a case where spins align only along a single axis (e.g., z).

iii. In a transverse field Ising model, the system is always ferromagnetic for any field strength.

iv. The Jordan-Wigner transformation maps 1D spin systems to non-interacting bosons.

v. At the critical point $h = J$ in the transverse Ising model, the energy gap remains large.

4 Relativistic Quantum Mechanics (R)

Block R1: Lorentz Invariance

Number of correct statements: 4

i. Non-relativistic QM fails when kinetic energy is comparable to rest mass energy.

ii. Physical laws must be invariant under Lorentz transformations in relativistic theory.

iii. The spacetime interval ds^2 is an invariant scalar in Minkowski space.

iv. The Schrödinger equation is relativistically covariant because it is first-order in time.

v. Proper time τ is a Lorentz scalar representing time in the particle's rest frame.

Block R2: Metric and Four-Vectors

Number of correct statements: 4

i. The metric tensor $g_{\mu\nu}$ is used to raise and lower four-vector indices.

ii. Lorentz transformations must satisfy $\Lambda^T g \Lambda = g$.

iii. Common Minkowski signatures are $(1, -1, -1, -1)$ or $(-1, 1, 1, 1)$.

iv. The Minkowski metric in Cartesian coordinates must have non-zero off-diagonal terms.

v. The four-momentum p^μ is $(E/c, \mathbf{p})$.

Block R3: Dirac Equation

Number of correct statements: 3

i. The Dirac equation is first-order in both time and space derivatives.

ii. It was designed to ensure a positive-definite probability density.

iii. The Dirac equation naturally predicts spin-1/2 for the described particles.

iv. The α and β matrices in the Dirac Hamiltonian are 2×2 matrices.

v. The Dirac equation only applies to massless particles.

Block R4: Spinors and Solutions

Number of correct statements: 4

i. Dirac wavefunctions are four-component spinors.

ii. For $\mathbf{p} = 0$, there are two $+mc^2$ energy solutions and two $-mc^2$ solutions.

iii. In the non-relativistic limit, two spinor components are "large" and two are "small".

iv. Negative energy solutions can be ignored as they have no physical meaning in Dirac theory.

v. The Feynman-Stueckelberg interpretation treats negative energy states as antiparticles.

Block R5: Klein Paradox

Number of correct statements: 4

i. The Klein paradox involves high transmission through a barrier as the potential height increases.

ii. It occurs when the potential step V exceeds $E + mc^2$.

iii. The paradox is resolved by considering particle-antiparticle pair production at the step.

iv. In the Klein zone, the wavefunction inside the barrier becomes oscillatory.

v. Relativistic tunneling is identical to the exponential decay seen in Schrödinger tunneling.

Solutions

Fundamentals (F)

- **F1:** i, ii, iv, v (4)
- **F2:** i, iv, v (3)
- **F3:** i, ii, iv, v (4)
- **F4:** i, ii (2)
- **F5:** i, iii, iv (3)

Scattering (S)

- **S1:** i, ii, iii, iv (4) — *iv is True:* $\int |f|^2 d\Omega = \sigma_{tot}$.
- **S2:** i, ii, iii (3)
- **S3:** i, iii, iv (3)
- **S4:** i, ii, iii, iv (4)
- **S5:** i, ii (2) — *v is False:* $\sigma_{k\to 0} = 4\pi R^2$.

Many Body (M)

- **M1:** ii, iii, v (3)
- **M2:** i, ii, iii, iv (4)
- **M3:** i, iv, v (3)
- **M4:** i, ii, iii, iv (4) — *iv is True: Hubbard* $\to$ *Heisenberg at* $U \gg t$.
- **M5:** i, ii (2)

Relativistic (R)

- **R1:** i, ii, iii, v (4) — *iv is False: Schrödinger is not covariant.*
- **R2:** i, ii, iii, v (4) — *iv is False:* $g_{\mu\nu}$ *is diagonal.*
- **R3:** i, ii, iii (3) — *iv is False* (4×4 *matrices), v is False (Massive particles).*
- **R4:** i, ii, iii, v (4) — *iv is False: Negative energy is essential.*
- **R5:** i, ii, iii, iv (4) — *v is False: Different from non-relativistic tunneling.*